Chapter 5: Introduction to Factorial Design

Many experiments involve the study of **the effect of two or more factors**. In general, **factorial designs** are **most efficient** for this type of experiment.

In a factorial design **all possible combinations** of the levels of the factors are investigated in each complete trial or replication of the experiment. For example, If there are a levels of factor A and b levels of factor B, each replicate contains all ab treatment combinations.

In this chapter, we will discuss about factorial designs.

Example 1: Consider the following data:

Response (y)	Factors A	Factors B
20	Low (-)	Low (-)
30	Low (-)	High (+)
40	High (+)	Low (-)
50	High (+)	High (+)

Define:

$$x_1 = \begin{cases} 1 & \text{when A is high} \\ -1 & \text{when A is low} \end{cases} \quad \text{and} \quad x_2 = \begin{cases} 1 & \text{when B is high} \\ -1 & \text{when B is low} \end{cases}$$

Therefore,

Response (y)	x_1	x_2
20	-1	-1
30	-1	$+1$
40	$+1$	-1
50	$+1$	$+1$

R Codes to Find the Estimate of Regression Model:

```
> y = c(20,30,40,50)
> x1= c(-1,-1,1,1)
> x2= c(-1,+1,-1,+1)
> model = lm(y~x1+x2+x1*x2)
> model
```

Call:
lm(formula = y ~ x1 + x2 + x1 * x2)

Coefficients:

(Intercept)	x1	x2	x1:x2
3.500e+01	1.000e+01	5.000e+00	-8.882e-16

Example 1: Consider the following data:

Response (y)	x_1	x_2
20	-1	-1
30	-1	$+1$
40	$+1$	-1
50	$+1$	$+1$

Consider the following regression model:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_1 x_{i2} + \beta_{12} x_{i1} x_{i2} + \varepsilon_i$$

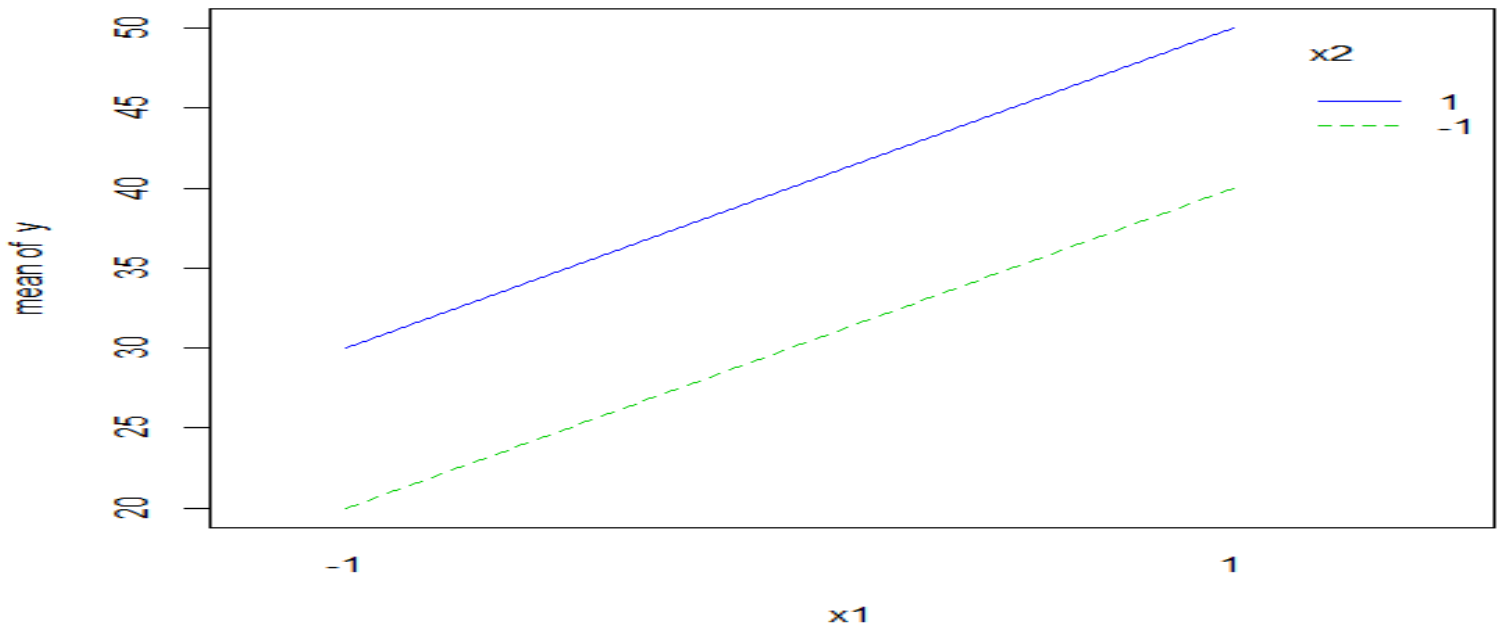
The estimate of the above regression model is:

$$\hat{y}_i = 35 + 10x_{i1} + 5x_{i2} + \mathbf{0}\, x_{i1} x_{i2}$$

Question: Why the estimate of β_{12} is zero?

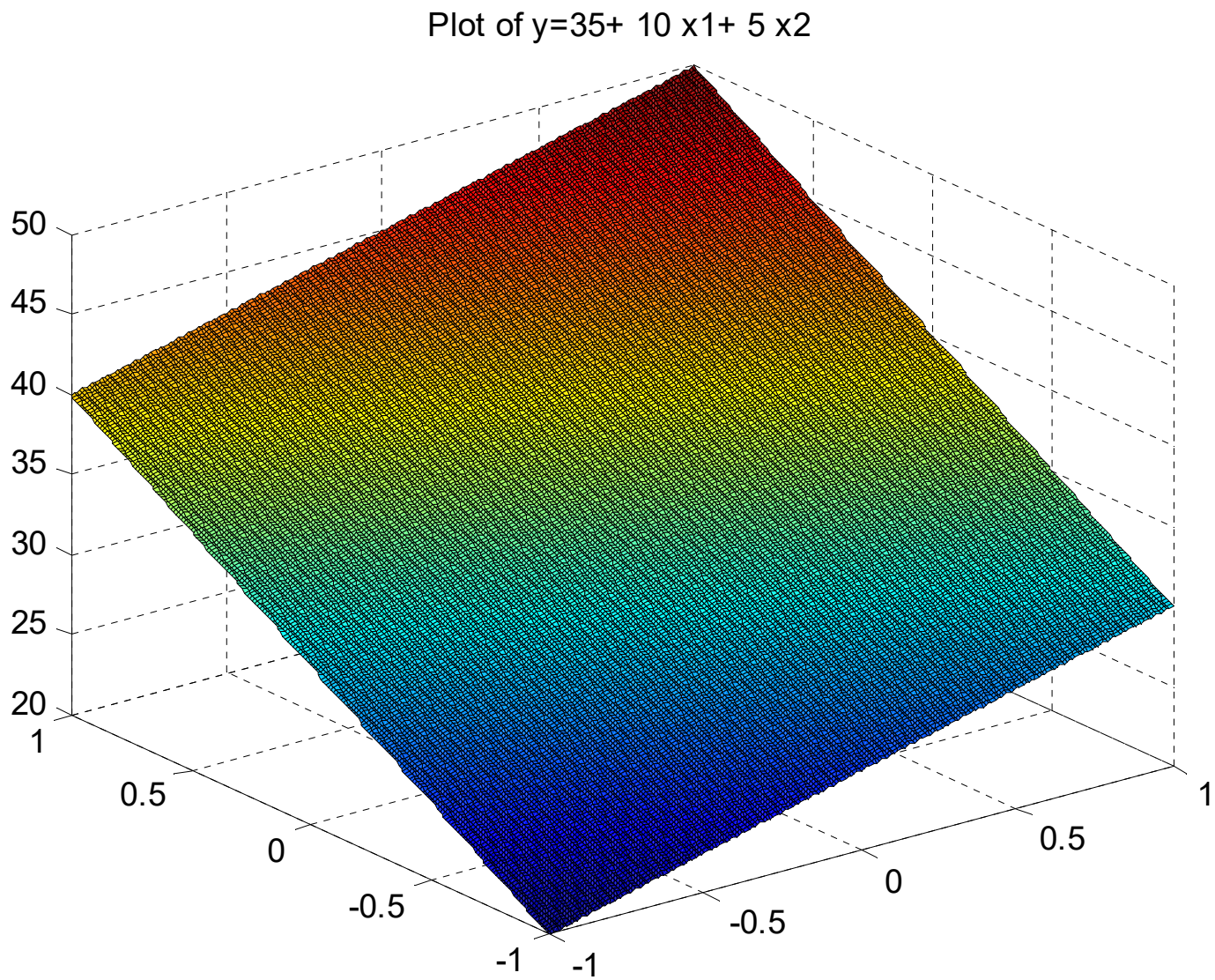
R Codes to Find Interaction (or Profile) plots:

```
> y = c(20,30,40,50)
> x1= factor(c(-1,-1,1,1))
> x2= factor(c(-1,+1,-1,+1))
> interaction.plot(x1,x2,y, col=c(3,4))
```



Example 1: The estimate of the above regression model is:

$$\hat{y}_i = 35 + 10x_{i1} + 5x_{i2} + \mathbf{0}\, x_{i1}x_{i2}$$



Example 1: Consider the following data:

Response (y)	Factors A	Factors B
20	Low (-)	Low (-)
30	Low (-)	High (+)
40	High (+)	Low (-)
50	High (+)	High (+)

At **low** level of **B**, the **effect** of **A** is 40 – 20 = 20
At **high** level of **B**, the **effect** of **A** is 50 – 30 = 20
The **overall** effect of **A** is: (20+20)/2 =20

The **interaction effect** is: (20 -20)/2 = **0**

Similarly,
At **low** level of **A**, the **effect** of **B** is: 30 – 20 = 10
At **high** level of **A**, the effect of B is: 50 - 40 = 10
The **overall** effect of **B** is: (10+10)/2 =10
The **interaction** effect is: (10 -10)/2 = **0**

Example 2: Consider the following data:

Response (y)	Factors A	Factors B
20	Low (-)	Low (-)
40	Low (-)	High (+)
50	High (+)	Low (-)
12	High (+)	High (+)

Define:

$$x_1 = \begin{cases} 1 & \text{when A is high} \\ -1 & \text{when A is low} \end{cases} \quad \text{and} \quad x_2 = \begin{cases} 1 & \text{when B is high} \\ -1 & \text{when B is low} \end{cases}$$

Therefore,

Response (y)	x_1	x_2
20	-1	-1
40	-1	$+1$
50	$+1$	-1
12	$+1$	$+1$

R Codes to Find the Estimate of Regression Model:

```
> y = c(20,40,50,12)
> x1= c(-1,-1,1,1)
> x2= c(-1,+1,-1,+1)
> model = lm(y~x1+x2+x1*x2)
> model
```

Call:
lm(formula = y ~ x1 + x2 + x1 * x2)

Coefficients:

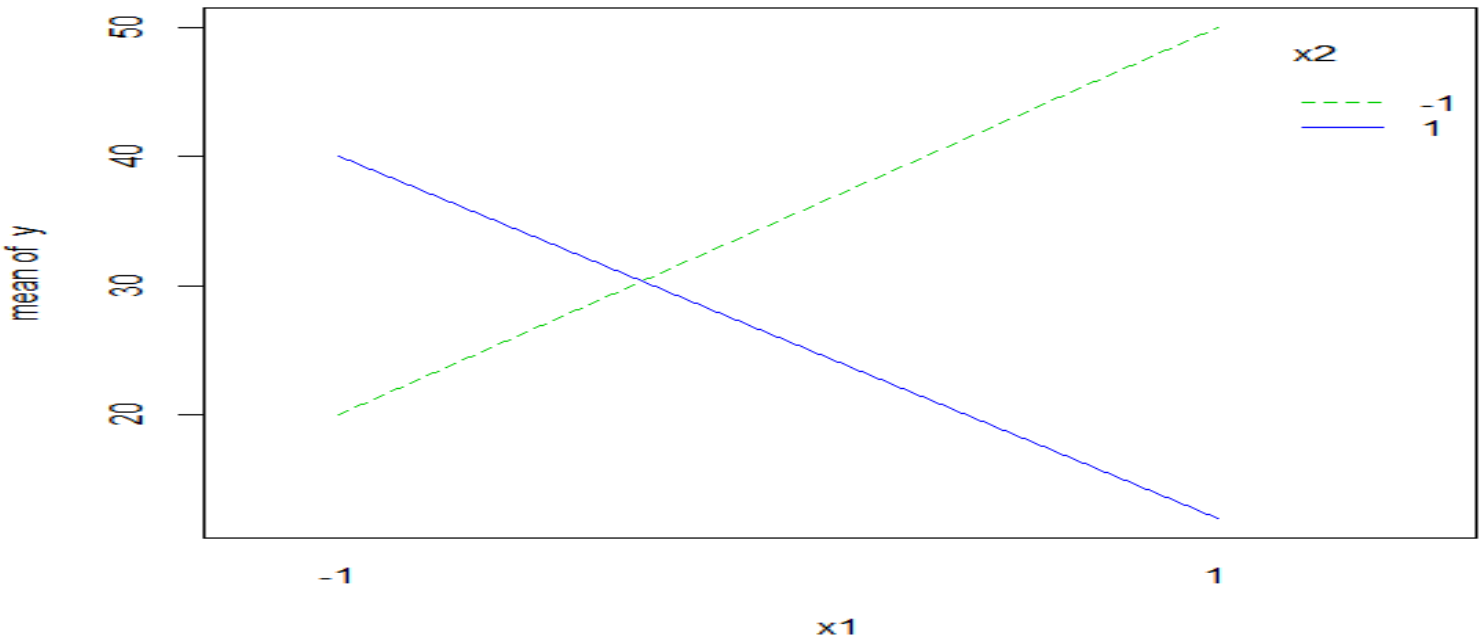
(Intercept)	x1	x2	x1:x2
30.5	0.5	-4.5	-14.5

The estimate of the regression model is:

$$\hat{y}_i = 30.5 + 0.5x_{i1} - 4.5x_{i2} - \mathbf{14.5}\, x_{i1}x_{i2}$$

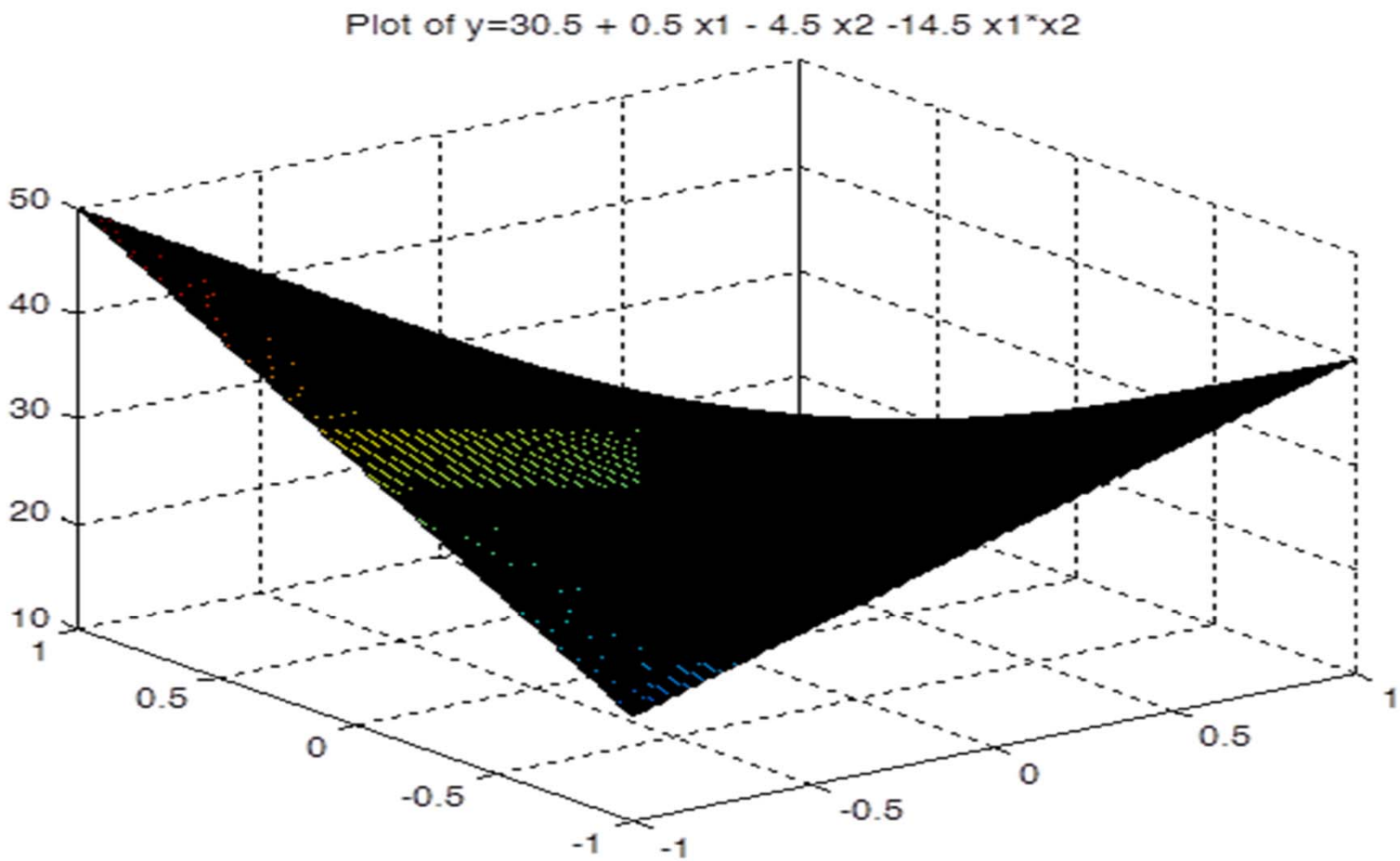
R Codes to Find Interaction (or Profile) plots:

```
> y = c(20,40,50,12)
> x1= factor(c(-1,-1,1,1))
> x2= factor(c(-1,+1,-1,+1))
> interaction.plot(x1,x2,y, col=c(3,4))
```



Example 2: The estimate of the above regression model is:

$$\hat{y}_i = 30.5 + 0.5x_{i1} - 4.5x_{i2} \mathbf{- 14.5}\, x_{i1}x_{i2}$$



Example 2: Consider the following data:

Response (y)	Factors A	Factors B
20	Low (-)	Low (-)
40	Low (-)	High (+)
50	High (+)	Low (-)
12	High (+)	High (+)

At **low** level of **B**, the **effect** of **A** is 50 – 20 = 30
At **high** level of **B**, the **effect** of **A** is 12 – 40 = -28
The **overall** effect of **A** is: (30+(-28))/2 =1

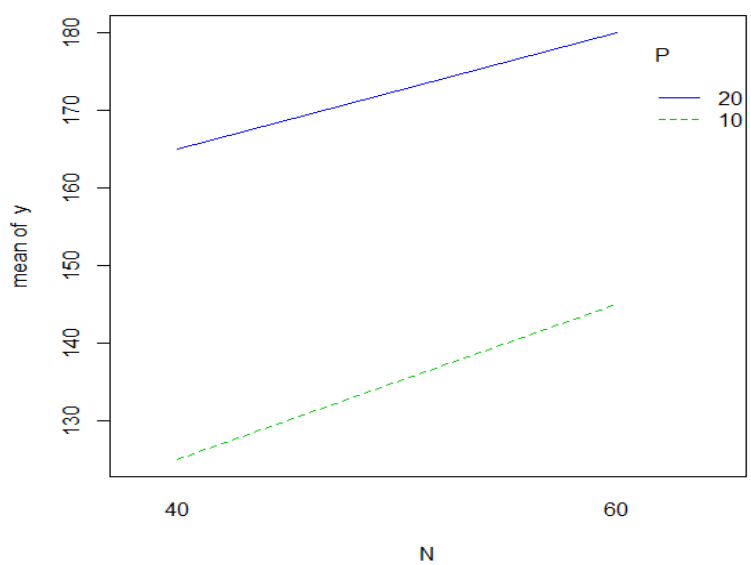
The **interaction effect** is: (-28 -30)/2 = **-29**

Similarly,
At **low** level of **A**, the **effect** of **B** is: 40 – 20 = 20
At **high** level of **A**, the effect of B is: 12 - 50 = -38
The **overall** effect of **B** is: (20+(-38))/2 =-9
The **interaction** effect is: (-38 -20)/2 = **-29**

Two factors A and B are said to **interact** if the difference in mean responses for two levels of one factor is not constant across levels of the second factor.

Example 3: Suppose we are interested in examining the **effect** of two levels of nitrogen at 40 and 60 pounds per plot and two levels of phosphorus at 10 and 20 pounds per plot on the yield of a crop. The following table gives the results of a completely randomized design where the four factor-level combinations are assigned at random to the experimental units.

Combination	Average Yield	Nitrogen	Phosphorus
1	145	60	10
2	125	40	10
3	165	40	20
4	180	60	20



Example 4 (example5.3.1 of textbook): An engineer is designing a battery for use in a device that will be subjected to some extreme variations in temperature. The only design parameter that he can select at this point is the plate material for the battery, and he has **three** possible choices. When the device is manufactured and is shipped to the field, the engineer has no control over the temperature extremes that the device will encounter, and he knows from experience that temperature will probably impact the effective battery life. However, temperature can be controlled in the product development laboratory for the purpose of a test.

The engineer decides to test all **three plate** materials at **three temperature** levels of 15^oF, 70^oF, and 125^oF -as these temperature levels are consistent with the product end-use environment. **Four batteries** are tested at each combination of plate material and temperature, and all 36 tests ate run in a **random order**. The observed battery lives are given in the table on the next page.

R Codes to Create a Two-Factor Factorial Design:

A factorial design can be easily created using R in several ways. One method is to use the bas R function **expand.grid**. This function creates a data frame containing all possible combinations of two factors.

$(\text{Levels})^{\text{Factors}}$

```
> D = expand.grid(Type=c(1,2,3), Temp = c(15,70,125))
```

The above code creates an un-replicated 3^2 factorial design in factor **material type = Type** and **temperature =Temp** with supplied levels for these factors.

To replicate every run in the design, the function **rbind** (which stacks one copy of the 3^2 factorial design on the top of the other.


```
> D= rbind(D,D,D,D)
```

R Codes to Create a Two-Factor Factorial Design:

```
> D= rbind(D,D,D,D)
```

To randomized the order of the run, the sample function can be used to create a random order of the run numbers 1 to 18. Then, the rows in the data frame D are ordered by this random order list.

```
> set.seed(2591)

> D = D[order(sample(1:36)),]

> plan = D[c("Type", "Temp")]

> write.csv(plan, file="factorial_plan.csv", row.names=FALSE)
```

Example 4:

Material Type	Temperature ( ) 15		70		125	
1	130	155	34	40	20	70
	74	180	80	75	82	58
2	150	188	136	122	25	70
	159	126	106	115	58	45
3	138	110	174	120	96	104
	168	160	150	139	82	60

Statistical Model: For $\mathrm{k} = 1,2,\ldots,\mathrm{n}$ and

$i = 1,2,\ldots,\mathrm{a} = \text{Levels of factor A}$, $\mathrm{j} = 1,2,\ldots,\mathrm{b} = \text{Levels of factor B}$, let

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

- $\hat{\mu} = \bar{y}_{...}$
- $\hat{\tau}_i = \bar{y}_{i..} - \bar{y}_{...}$
- $\hat{\beta}_j = \bar{y}_{.j.} - \bar{y}_{...}$
- $\widehat{(\tau\beta)}_j = \bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...}$

Statistical Model:

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

Define

$$SS_{Total} = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n}(y_{ijk} - \bar{y}_{...})^2 ,$$

$$SS_A = nb\sum_{i=1}^{a}(\bar{y}_{i..} - \bar{y}_{...})^2 ,$$

$$SS_B = na\sum_{j=1}^{b}(\bar{y}_{.j.} - \bar{y}_{...})^2 ,$$

$$SS_{AB} = n\sum_{i=1}^{a}\sum_{j=1}^{b}(\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...})^2 ,$$

$$SS_{Error} = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n}(y_{ijk} - \bar{y}_{ij.})^2,$$

Statistical Model:

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

It can be shown that

$$SS_{Total} = SS_A + SS_B + SS_{AB} + SS_{Error}$$

$$E(MS_A) = \sigma^2 + \frac{nb\sum_{i=1}^{a}\tau_i^2}{a-1}, \quad E(MS_B) = \sigma^2 + \frac{na\sum_{j=1}^{b}\beta_j^2}{b-1}$$

$$E(MS_{AB}) = \sigma^2 + \frac{n\sum_{i=1}^{a}\sum_{j=1}^{b}(\tau\beta)_{ij}^2}{(a-1)(b-1)}, \quad E(MS_{Error}) = \sigma^2$$

The **ANOVA table**:

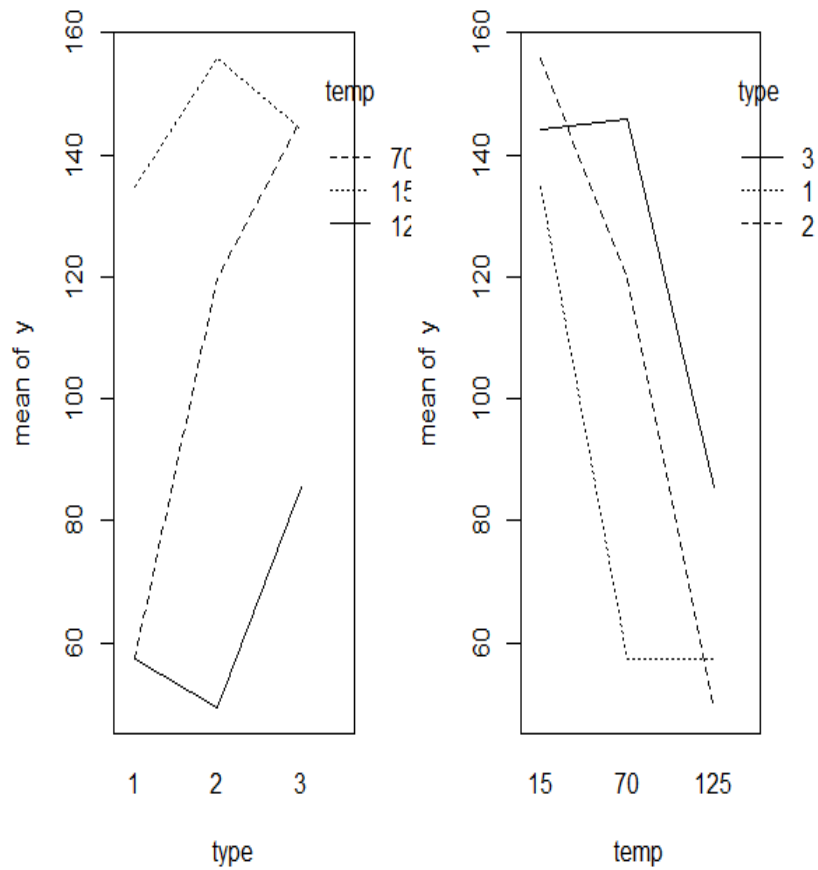
Source	df	SS	MS	F
A	$a-1$	SSA	$\mathrm{MSTr} = \frac{\mathrm{SS}A}{a-1}$	$F = \frac{\mathrm{MS}A}{\mathrm{MSE}}$
B	$b-1$	SSB	$\mathrm{MSB} = \frac{\mathrm{SS}B}{b-1}$	$F = \frac{\mathrm{MS}B}{\mathrm{MSE}}$
AB	$(a-1)(b-1)$	SAB	$\mathrm{MSE} = \frac{\mathrm{SS}AB}{(a-1)(b-1)}$	$F = \frac{\mathrm{MS}AB}{\mathrm{MSE}}$
Error	$ab(n-1)$	SSE	$\mathrm{MSE} = \frac{\mathrm{SSE}}{ab(n-1)}$	
Total	$nab-1$	SST		

R Codes to Obtain Profile plots:

```
> y <- c(130, 155,  74, 180,  34,  40,  80,  75,  20,  70,  82,  58, 150,
        188, 159, 126, 136, 122, 106, 115,  25,  70,  58,  45, 138,
        110, 168,160, 174, 120, 150, 139,  96, 104,  82,  60)

> type <- as.factor(rep(1:3,each=12))
> temp <- as.factor(rep(c(15,70,125),
                        each=4,times=3))
> data <- data.frame(type, temp, y)

> par(mfrow=c(1,2))
> interaction.plot(type,temp,y)
> interaction.plot(temp,type,y)
```



R Codes to Obtain ANOVA Table for the Factorial Design:

```
> model <- lm(y~type+temp+type*temp)
> anova(model)
```

Analysis of Variance Table

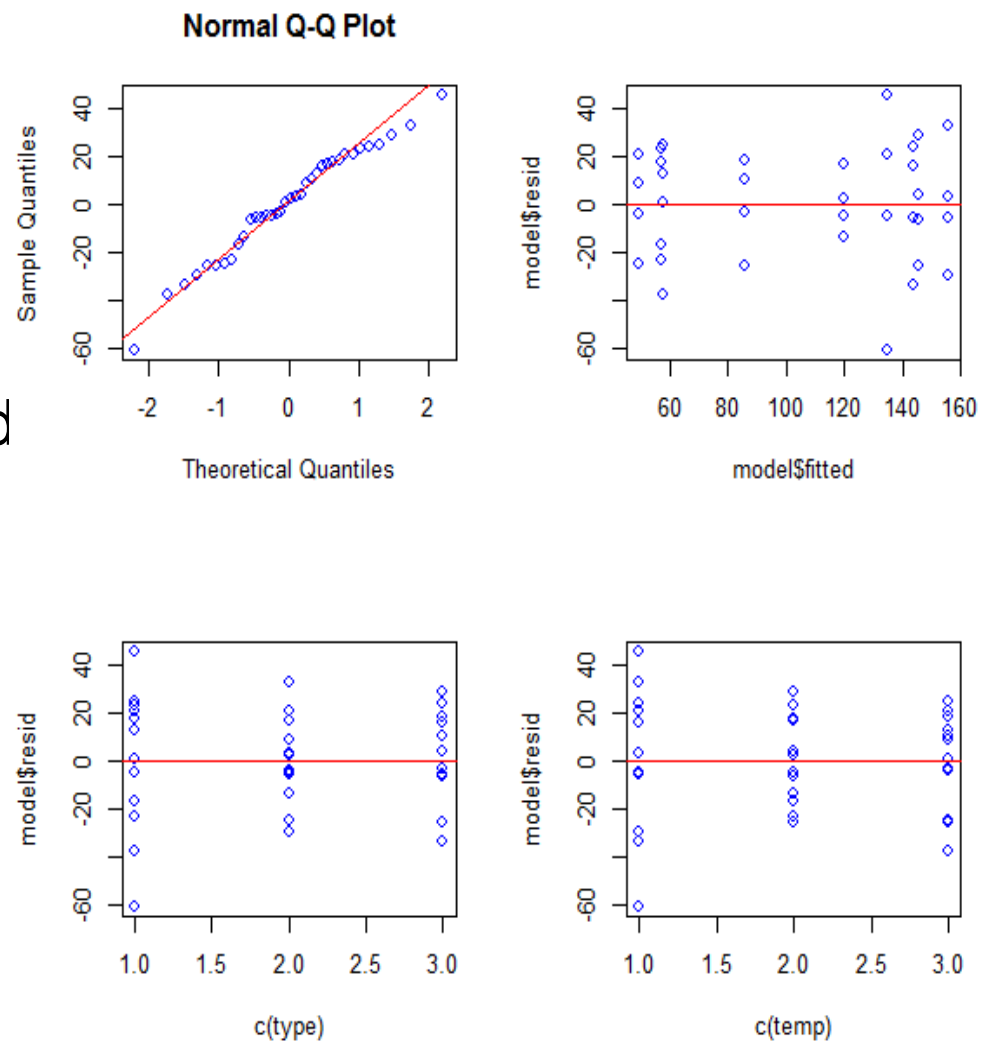
Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
type	2	10684	5341.9	7.9114	0.001976 **
temp	2	39119	19559.4	28.9677	1.909e-07 ***
type:temp	4	9614	2403.4	3.5595	0.018611 *
Residuals	27	18231	675.2		

Signif. codes: 0 ‘*’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1**

R Codes to Check Assumptions:

```
> windows()
> par(mfrow=c(2,2))
> qqnorm(model$resid,
    col="blue")
> qqline(model$resid, col=2)
> plot(model$fitted,model$resid
    col="blue")
> abline(h=0, col="red")
> plot(c(type), model$resid,
        col="blue")
> abline(h=0, col="red")
>plot(c(temp), model$resid,
        col="blue")
>abline(h=0, col="red")
```



Example 4: As suspected, the interaction effects are quite significant.

Therefore, There is **no** battery type which is '**best**' at all temperatures.

If interactions were **NOT** significant one could compare the $\mu + \tau_i$ by seeing which of the differences $\hat{\mu} + \hat{\tau}_i = \bar{y}_{i..}$ were significantly different.

Since there are interaction effects, we can only make comparisons at fixed levels of the other factor.

For instance when temperature is 70,($j = 2$) we can compare the means

$$\mu_{i2} = \mu + \tau_i + \beta_2 + (\boldsymbol{\tau\beta})_{\boldsymbol{i2}}$$

An estimate of μ_{i2} is $\bar{y}_{i2.}$, an average of n observations.

Example 4: As suspected, the interaction effects are quite significant.

$$\mu_{i2} = \mu + \tau_i + \beta_2 + (\boldsymbol{\tau\beta})_{\boldsymbol{i2}}$$

An estimate of μ_{i2} is $\bar{y}_{i2.}$, an average of n observations.

The 95% Tukey's confidence intervals for $\mu_{i2} - \mu_{j2}$ are:

$$\bar{y}_{i2.} - \bar{y}_{j2.} \pm \frac{qtukey(0.95,3,27)}{\sqrt{2}}\left(\sqrt{MSE}\sqrt{\frac{1}{n}+\frac{1}{n}}\right)$$

Note: $\bar{y}_{12.} = 57.25$, $\bar{y}_{22.} = 119.75$, $\bar{y}_{32.} = 145.75$,

$$ME = \frac{3.506426}{\sqrt{2}}\left(\sqrt{675.2}\sqrt{\frac{1}{4}+\frac{1}{4}}\right) = 45.56$$

$$\mu_{12} - \mu_{22}: 57.25 - 119.75 \pm 45.56 = -62.50 \pm 45.56$$
$$\mu_{12} - \mu_{32}: 57.25 - 145.75 \pm 45.56 = -88.50 \pm 45.56$$
$$\mu_{22} - \mu_{32}: 119.75 - 145.75 \pm 45.56 = -26.00 \pm 45.56$$

We conclude that μ_{12} is significantly smaller than μ_{22} and μ_{32}, but these two are not significantly different from each other.

R Codes to Obtain Averages:

You may write a short program, such as the following, to obtain averages:

```
> means <- matrix(nrow=3,ncol=3)
   for(i in 1:3) {
    for (j in 1:3)
       means[i,j]=  mean(y[type==i & temp == c(15,70,125)[j]])}
 > means
```

	[,1]	[,2]	[,3]
[1,]	134.75	57.25	57.5
[2,]	155.75	119.75	49.5
[3,]	144.00	145.75	85.5

R Codes to Obtain Averages:

You may also use **model.tables** function to the averages.

```
> library(daewr)
> model <- aov(y~type+temp+type*temp)
> summary(model)
> model.tables(model, type="means", se=T)
```

The **mode.tables** produces the results shown in the next page.

The top line is the estimate of the overall mean, $\hat{\mu}$.

The next two sections show the marginal means for each factor along with the standard deviation of the values average in each mean.

The next section shows all means, and the final section shows the standard errors of the difference in marginal and cell means.

Outputs of model.tables(model, type="mean", sd=T):

```
Tables of means
Grand mean
   105.5278
type
     1      2      3
 83.17 108.33 125.08
 temp
    15     70    125
144.83 107.58  64.17
type:temp
                 temp
type    15    70    125
   1  134.75   57.25   57.50
   2  155.75  119.75   49.50
   3  144.00  145.75   85.50
```

Standard errors for differences of means

	type	temp	type:temp
	10.61	10.61	18.37
replic.	12	12	4

Test for Interaction with One Replicate per Cell: Sometimes we can make only **one observation** per cell ($\boldsymbol{n = 1}$). Then, for example $\bar{y}_{ij1} = \bar{y}_{ij.}$. So

$$SS_{Error} = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n}(y_{ijk} - \bar{y}_{ij.})^2 = 0$$

Recall that:

$$SS_{AB} = n\sum_{i=1}^{a}\sum_{j=1}^{b}(\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...})^2\,,$$

For $n = 1$, we have

$$SS_{AB} = n\sum_{i=1}^{a}\sum_{j=1}^{b}(\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...})^2$$

$$= \sum_{i=1}^{a}\sum_{j=1}^{b}(y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..})^2 = SS_{Error}$$

Therefore, the interaction SS for $n = 1$ is, in fact, SS for error.

Example 5: The data in the next page table is a portion of data from a study to validate an assay of viral contamination (See Lin, L. I. and Stephenson, W. R. (1998), "*Validating an assay of viral contamination*" in R. Peck, L. D. Haugh, and A. Goodman, editors, Statistical Case Studies: *A Collaboration Between Academe and Industry*, pages 43-48, SIAM, Philadelphia).

Assay of viral contamination are used to determine the presence (and amount) of a specific virus in biological products such as blood clotting Factor Eight.

An experiment, or run, consists of making a solution with a known viral contamination, allowing the virus in a contaminated solution to grow, then measuring the result.

The experimental unit is the specific viral sample in combination with the place and the time where it is allowed to grow.

Example 5:

Sample

Dilution

	1	2	3	4	5	6
3	1.87506	1.74036	1.79934	2.02119	1.79934	1.59106
4	1.38021	1.36173	1.25527	1.39794	1.20412	1.25527
5	0.60206	0.90309	0.95424	1.00000	0.60206	0.60206

Factor **A** represent the sample number, or solution with which the viral sample is mixed (or spiked).

Factor **B** represent different dilutions of the spiked sample.

The measured **response** is log_{10} of the plaque forming units per mL of solution.

Note: Factor A is not a quantitative factor.

```
> y = c( 1.87506, 1.74036, 1.79934, 2.02119, 1.79934, 1.59106,
         1.38021, 1.36173, 1.25527, 1.39794, 1.20412, 1.25527,
         0.60206, 0.90309, 0.95424, 1.00000, 0.60206, 0.60206)
> A = factor(rep(1:6, times =3))    > B=  factor(rep(3:5, each =6))
> model = lm(y~A+B+A:B)
> anova(model)
Analysis of Variance Table
Response: y
          Df    Sum Sq    Mean Sq    F value    Pr(>F)
A          5    0.1948    0.03896
B          2    3.1664    1.58319
A:B       10    0.1283    0.01283
Residuals  0    0.0000
Warning message:
In anova.lm(model) :
  ANOVA F-tests on an essentially perfect fit are unreliable
```

There is still however a way to test for interactions, if we assume that they take a simple form:

$$(\tau\beta)_{ij} = \gamma\tau_i\beta_j$$

We carry out '**Tukey's one d.f. test for interaction**', which is an application of the usual 'reduction in SS' hypothesis testing principle.

The '**full**' model is: $y_{ij} = \mu + \tau_i + \beta_j + \gamma\tau_i\beta_j + \varepsilon_{ij}$

Under the null hypothesis $\boldsymbol{H_0: \gamma = 0}$ of no interactions, the '**reduced**' model is: $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$

The F-statistic is: $F = \frac{SS_{Reduced} - SS_{Full}}{MS_{Full}} \sim F^{1}_{(a-1)(b-1)-1}$

The difference $SS_N = SS_{Reduced} - SS_{Full}$ is called the "SS for non-additivity" and uses one degrees of freedom to estimate one parameter γ.

The **ANOVA table**:

Source	df	SS	MS	F
A	$a-1$	SSA	$\mathrm{MSA} = \dfrac{\mathrm{SS}A}{a-1}$	$\mathrm{F} = \dfrac{\mathrm{MS}A}{\mathrm{MSE}}$
B	$b-1$	SSB	$\mathrm{MSB} = \dfrac{\mathrm{SS}B}{b-1}$	$\mathrm{F} = \dfrac{\mathrm{MS}B}{\mathrm{MSE}}$
N	1	SSN	$\mathrm{MSN} = \dfrac{\mathrm{SS}N}{1}$	$\mathrm{F} = \dfrac{\mathrm{MS}N}{\mathrm{MSE}}$
Error	$(a-1)(b-1)-1$	SSE	$\mathrm{MSE} = \dfrac{\mathrm{SSE}}{(a-1)(b-1)-1}$	
Total	$ab-1$	SST		

Note: It turns out that

$$SS_N = \frac{ab\{\sum_{ij} y_{ij}\bar{y}_{i.}\,\bar{y}_{.j} - (SS_A + SS_B + ab\bar{y}_{..}^2)\bar{y}_{..}\}^2}{SS_A \times SS_B}$$

Then SS_{Error} which is, in fact, SS_{Full} can be obtained by

$$SS_{Error} = SS_{Reduced} - SS_N$$

The function **Tukey1df** in the R package **daewr** calculate the non-additivity sum of squares and prints a report. The first column in the data frame used by this function is a numeric response, the second column is the indicator for factor A, and the third column is the indicator for factor B.

```
> library(daewr)
>Tukey1df(virus) #Use help(Tukey1df) to know about data frame
```

Source	df	SS	MS	F	Pr>F
A	5	0.1948	0.039		
B	2	3.1664	1.5832		
Error	10	0.1283	0.0513		
NonAdditivity	1	0.0069	0.0069	0.51	0.4932
Residual	9	0.1214	0.0135		

In the results, It can be seen that the interaction (or non-additivity) is not significant. Therefore, for this data, it would be appropriate to fit the additive model $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$.

R Codes to Obtain ANOVA Tables for Virus Data:

```
> model2 = lm(y~A+B)
> anova(model2)
```

Analysis of Variance Table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
A	5	0.1948	0.03896	3.0377	0.06349 .
B	2	3.1664	1.58319	123.4445	8.939e-08 ***
Residuals	10	0.1283	0.01283		

Factorial Designs with Multiple Factors (CRFD): Two-factor factorial design are more efficient than studying each factor separately in one-factor design.

Likewise, when there are many factors are under study, it is more efficient to study them together in a multifactor factorial design than it is to study them separately in groups of two using two-factor factorial design.

In a multifactor factorial not only is it possible to detect two-factor interactions between any pair of factors, but it is also possible to detect higher order interaction between groups of factors.

Full 3 factor model is:

$$y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_k$$
$$+(\tau\beta)_{ij} + (\tau\gamma)_{ik} + (\beta\gamma)_{jk}$$
$$+(\tau\beta\gamma)_{ijk} + \varepsilon_{ijkl}$$

The treatment combinations in a **multi-factor factorial** consists of **all possible combinations** of the levels of all factors.

A design can be produced using **expand.grid** function in R, or using the **gen.factorial** function in the **AlgDesign** package or using functions from other packages.

Example 6 (Example 5.3 from the textbook): A soft bottler is interested in obtaining more uniform fill heights in the bottles produced by his manufacturing process. The filling machine theoretically fills each bottle to the correct target height, but in practice, there is variation around this target, and the bottler would like to understand better the sources of this variability and eventually reduce it.

The process engineer can control three variables during the filling process, the percent carbonation (A), the operating pressure in the filter (B), and the bottles produced per minute or the line speed (C).

Example 6 (Cont.): The pressure and speed are easy to control, but the percent carbonation is more difficult to control during actual manufacturing because it varies with product temperature.

However, for purpose of an experiment, the engineer can control carbonation at three levels: 10, 12 and 14 percent.

He chooses two levels for pressure (25 and 30 psi) and two levels for line speed (200 and 250 bpm).

He decides to run two replicates of a factorial design in these three factors, with **all** 24 runs takes in **random order**.

The response variable observed is the average deviation from the target fill height observed in a production run of bottles at each set of conditions. The data that resulted from this experiment are shown in the table on the next page. Positive deviations are fill height above the target, whereas negative deviations are fill heights below the target.

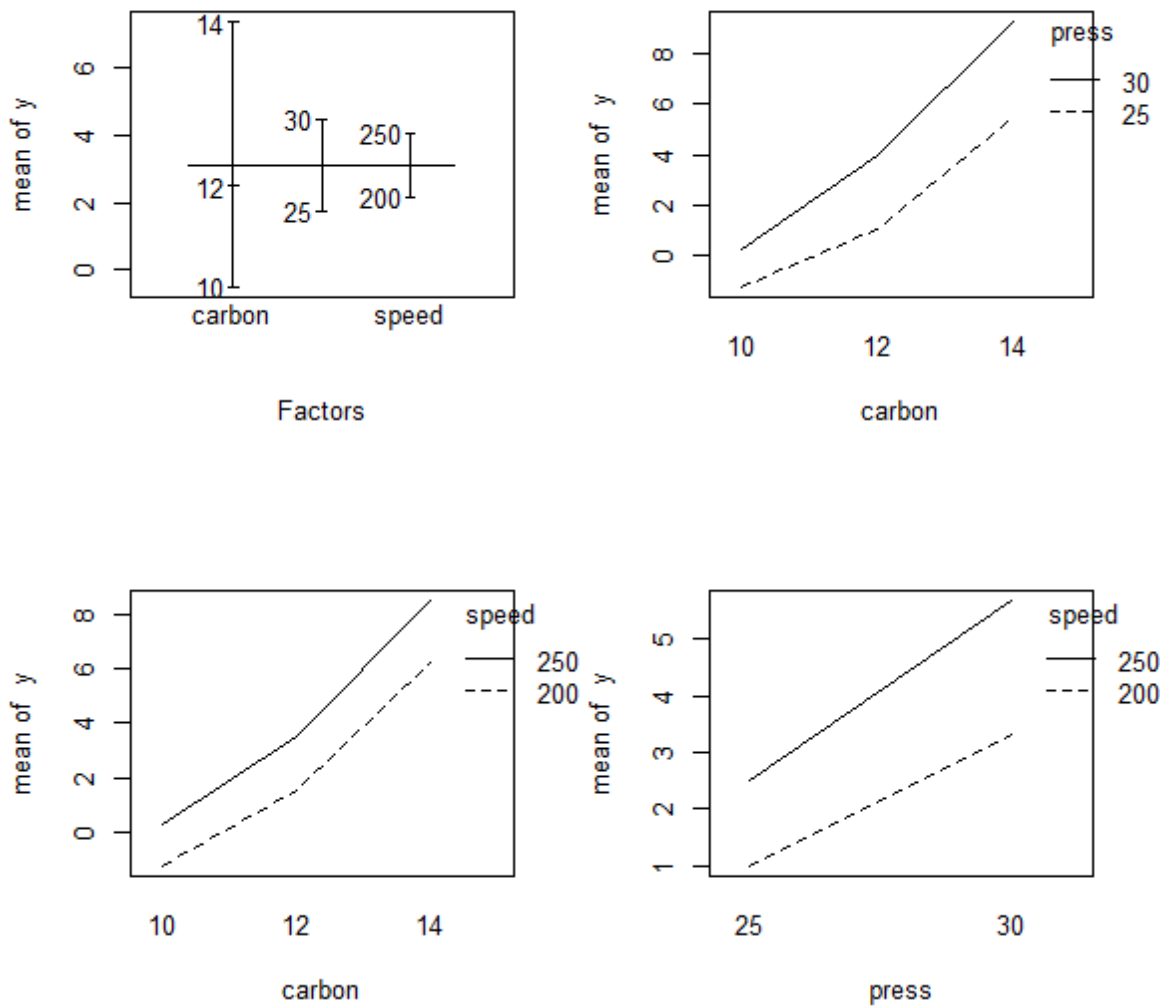
Example 6 (Cont.):

Percent Carbonation (A)	Operating Pressure (B) 25 psi — Line Speed (C) 200	250	30 psi — Line Speed (C) 200	250
10	-3	-1	-1	1
	-1	0	0	1
12	0	2	2	6
	1	1	3	5
14	5	7	7	10
	4	6	9	11

```
> y <- c(-3,-1,0,1,5,4, -1,0,2,1,7,6, -1,0,2,3,7,9, 1,1,6,5,10,11)
> carbon <- as.factor(rep(c(10,12,14),each=2, times=4))
> press <- as.factor(rep(c(25,30), each=12))
> speed <- as.factor(rep(c(200,250), each=6, times=2))
> data <- data.frame(y, carbon, press, speed)
> attach(data)
 >data
```

R Codes to Find Interaction Plot:

```
> par(mfrow=c(2,2))
> plot.design(data)
> interaction.plot(carbon,
                   press,y)
> interaction.plot(carbon,
                   speed,y)
> interaction.plot(press,
                   speed,y)
```



R Codes to Find ANOVA for CRFD:

```
> model1 <- lm(y ~ carbon + press + speed +
    carbon*press + carbon*speed + press*speed +  carbon*press*speed)
> anova(model1)
```

Analysis of Variance Table

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
carbon	2	252.750	126.375	178.4118	1.186e-09	***
press	1	45.375	45.375	64.0588	3.742e-06	***
speed	1	22.042	22.042	31.1176	0.0001202	***
carbon:press	2	5.250	2.625	3.7059	0.0558081	.
carbon:speed	2	0.583	0.292	0.4118	0.6714939	
press:speed	1	1.042	1.042	1.4706	0.2485867	
carbon:press:speed	2	1.083	0.542	0.7647	0.4868711	
Residuals	12	8.500	0.708			

R Codes to Find ANOVA for CRFD without Interaction:

```
> model2 <- lm(y ~ carbon + press + speed)
> anova(model2)
```

Analysis of Variance Table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
carbon	2	252.750	126.375	145.891	2.950e-12 ***
press	1	45.375	45.375	52.382	7.176e-07 ***
speed	1	22.042	22.042	25.446	7.201e-05 ***
Residuals	19	16.458	0.866		

Signif. codes: 0 '*' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1**

R Codes to Compare Two Models:

```
> anova(model2, model1)
```

Analysis of Variance Table

Model 1: y ~ carbon + press + speed

Model 2: y ~ carbon + press + speed + carbon * press + carbon * speed + press * speed + carbon * press * speed

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	19	16.458				
2	12	8.500	7	7.9583	1.605	0.2249

Example 7: Consider an example of a multi-factor factorial design in marketing research.

A company whose sales are made online through a Web page would like to increase the proportion of visitors to their Web site that sign up for their service by optimally configuring their Web page.

In order to buy from the company, customers must sign up and fill out a form supplying their email address along with other required fields. Once a customer sign up, the company has contact information for their database and can email advertisements, special offers and so forth.

The company would like to experiment by testing different configurations of their Web page to see if they can increase the number of visitors to their site that actually sign up.

Example 7 (Cont.):

The **experimental units** in this study will be **individuals** who visit the company Web site. The response is binary; the customer either sign up or does not.

The **factors** under study are characteristics that change the appearance of the Web page.

Factor **A** is the **background** alternatives for the page with three options.

Factor **B** is the **font size** in the main banner, with three levels

Factor **C** is the **text color** with two alternatives.

Factor **D** is a choice between a **sign-up button** or **link**.

What Have We Learned?

We discussed factorial design in this chapter. In summary, **factorial designs** have several **advantages**, such as:

- They are **more efficient** than one-factor-at-a-time experiments.
- A factorial design is necessary when interactions may be present **to avoid misleading conclusion**.
- Factorial designs allow the effects of a factor to be estimated at several levels of the other factors, yielding conclusions that are valid over a range of experimental conditions.